

THE SYNC QUESTION

From NVIDIA's G-Sync to AMD's FreeSync, what is Adaptive Sync technology all about?

Not all of us have the best possible rig when it comes to gaming (or sometimes even video playback). Nonetheless, it does get on our nerves when the low refresh rate on our displays causes the dreaded screen tearing and spoils the entire experience. Screen tearing occurs when a display ends up with information from more than one frame in a single shot. You can mostly notice it when the camera, in a game or a movie, pans sideways and the edges of objects fail to line up.

The main culprit for this is the disparity between the refresh rates of the display and the feed from the display driver. The one for the display is fixed by the manufacturer, whereas that of the driver keeps changing based on the computational load it is bearing. So, while the process of sending a frame to the monitor is ongoing at the GPU, the GPU itself switches to a new frame. You can notice this discontinuity, or tear travel down the screen as the difference in phase between the GPU and the monitor gradually reduces, but in some worse cases there can several tears at once.